TWEET SENTIMENT ANALYSIS USING NLP

Understanding Customer Opinions from Social Media

Group 6

BUSINESS PROBLEM

Titan Electronic Company sells high-end electronic products (iPhones, iPads) in a highly competitive market, where:

- Customer perception strongly influences sales and brand loyalty
- Customer feedback on X (twitter) is large and unstructured
- Difficult to analyse sentiments manually at scale

OBJECTIVE & KEY QUESTIONS

This project aims to build a sentiment classification system using NLP that can automatically analyze customer opinions from Twitter data

- The analysis is guided by the following key questions:
- How is sentiment distributed across tweets?
- How does data quality affect model performance?
- How well do different models classify sentiment?
- Which model best detects negative sentiment?

This sets the direction of the entire project.

DATASET & SOURCES

- The dataset consists of over 9,000 tweets containing customer opinions about products and brands. Each tweet includes text, associated product information, and a sentiment label.
- The data represents real-world, unstructured input, meaning it contains noise such as slang, abbreviations, and inconsistent formatting.
- This makes it suitable for testing real NLP techniques.

CLASS DISTRIBUTION

The dataset contains tweets classified into four sentiment categories. The distribution shows that the majority of tweets fall under neutral (no emotion toward brand or product), followed by relatively similar proportions of positive and negative sentiment.

A smaller portion of the data is labeled as “I can’t tell”, representing ambiguous or unclear sentiment.

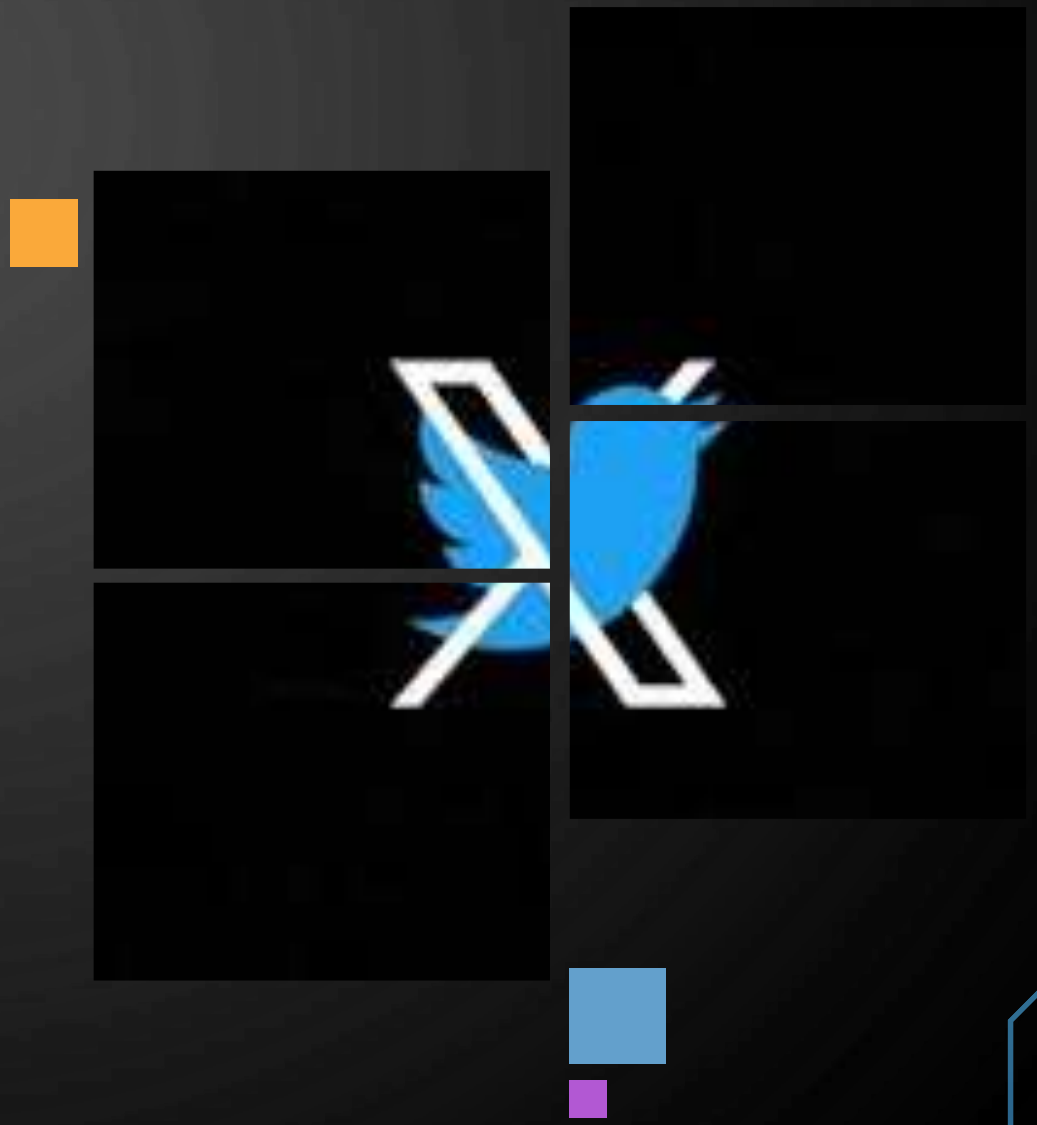
- The dataset is imbalanced model performance must prioritize detecting minority classes, especially negative sentiment.

Sentiment	Percentage
I can’t tell	17.628205
Negative emotion	18.746924
No emotion toward brand or product	17.43764
Positive emotion	18.183625

Sentiment	Number(s)
I can’t tell	5372
Negative emotion	2968
No emotion toward brand or product	569
Positive emotion	156

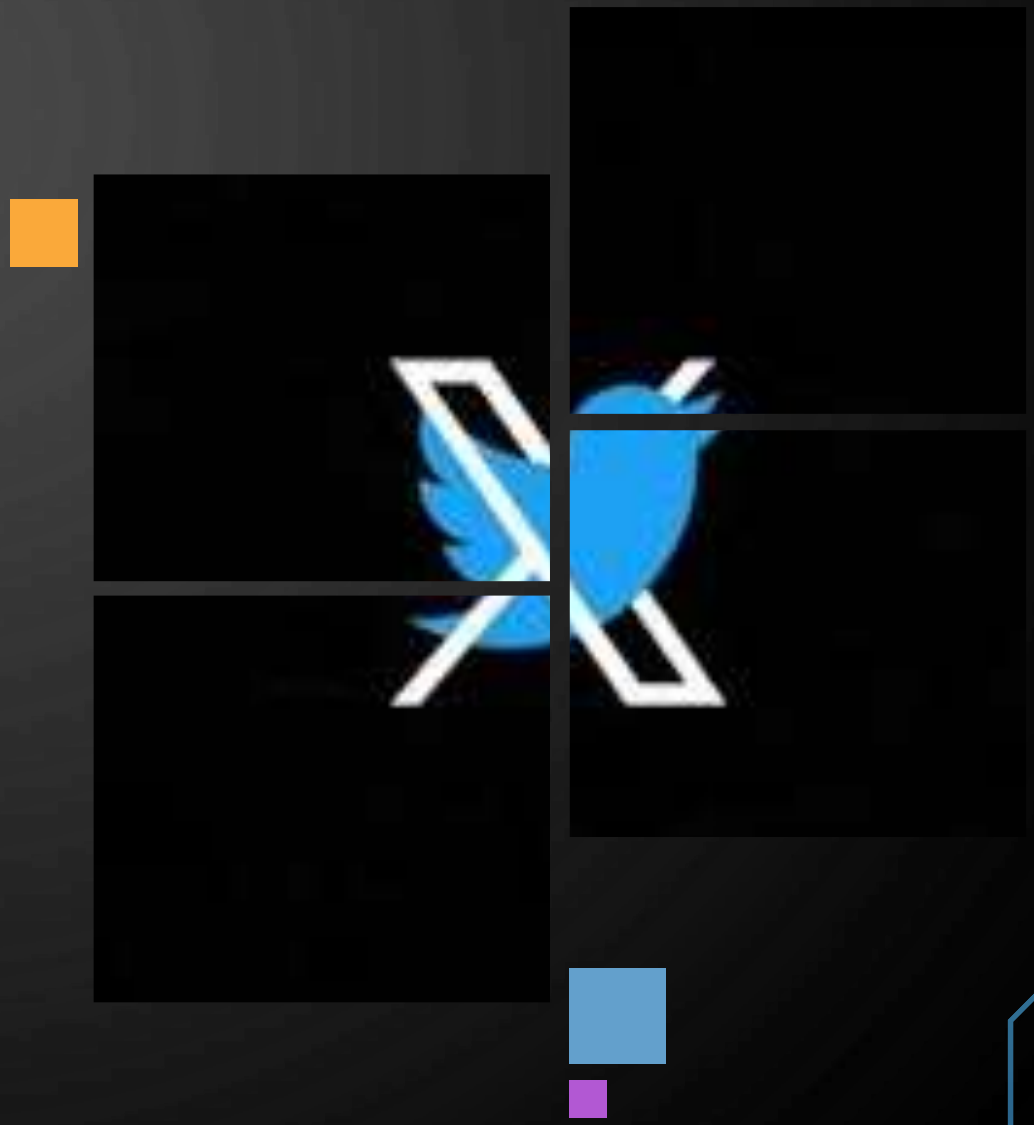
DATA PREPARATION

- To ensure data quality, several cleaning steps were applied. Missing tweet text entries were removed since they cannot be used in analysis. Duplicate tweets were also eliminated to avoid bias and repetition in model training.
- Ambiguous labels such as “I can’t tell” were excluded to maintain clear classification boundaries. Additionally, product names were standardized into broader categories such as Apple and Google.
- These steps improve consistency and ensure the model learns from meaningful patterns.



TEXT PROCESSING

- The raw tweet text was transformed through a structured preprocessing pipeline. This included converting text to lowercase, removing noise such as URLs and punctuation, and breaking sentences into tokens.
- Stopwords were removed to eliminate irrelevant words, and lemmatization was applied to reduce words to their base form.
- The result is a cleaned representation of text that retains meaning while removing unnecessary variation.



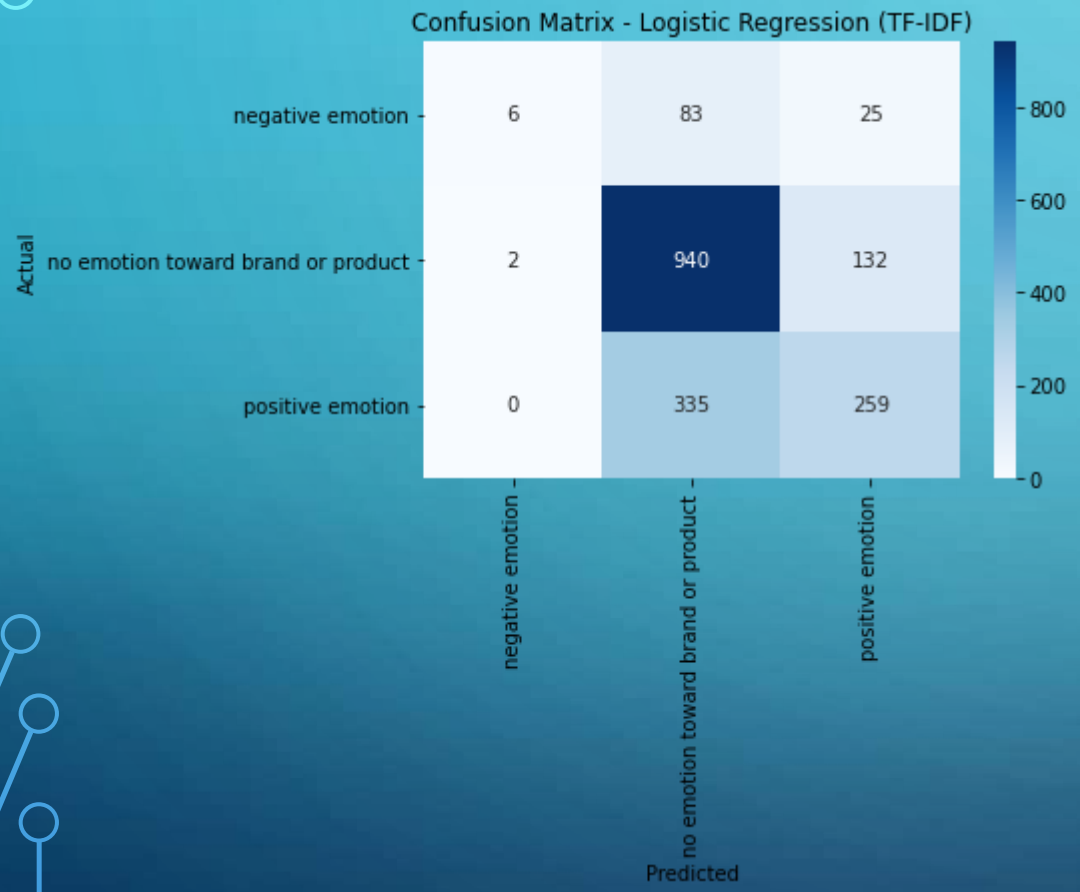
FEATURE ENGINEERING

- To prepare the data for machine learning, TF-IDF vectorization was used. This technique converts text into numerical values based on word importance.
- Words that appear frequently in a specific tweet but not across all tweets receive higher importance, allowing the model to focus on meaningful features.
- This step is essential because it enables algorithms to detect patterns in textual data.

MODELING APPROACH

- Four models were developed to evaluate performance under different configurations. Baseline models included Logistic Regression and Naive Bayes using standard TF-IDF features.
- Improved versions of these models introduced adjustments such as expanded feature ranges and class balancing techniques.
- The goal was to compare how simple and improved models handle imbalanced data and sentiment classification.

LOGISTIC REGRESSION (BASELINE)



Interpretation:

•Strong performance on neutral class

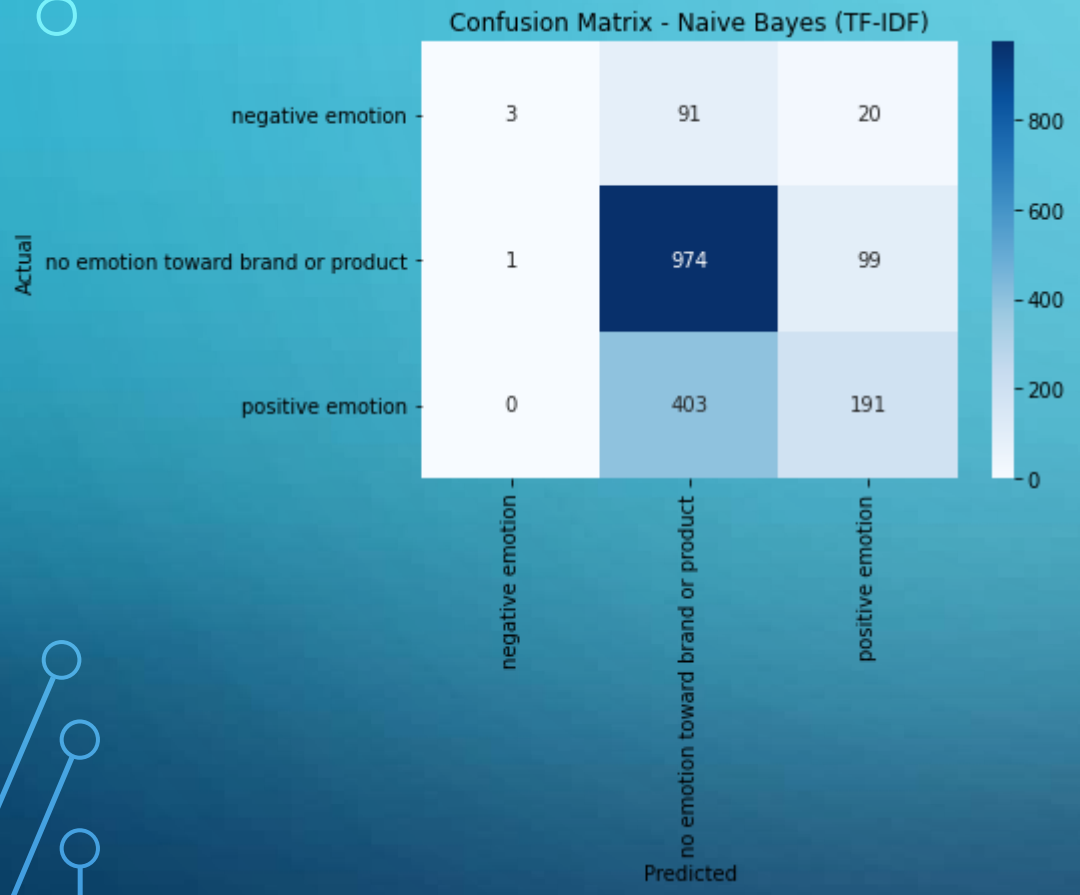
•Very low detection of negative tweets

•Predictions heavily biased toward majority class

Insight:

The model appears accurate but fails the main objective of identifying negative sentiment.

NAIVE BAYES (BASELINE)



Interpretation:

Extremely high bias toward neutral predictions

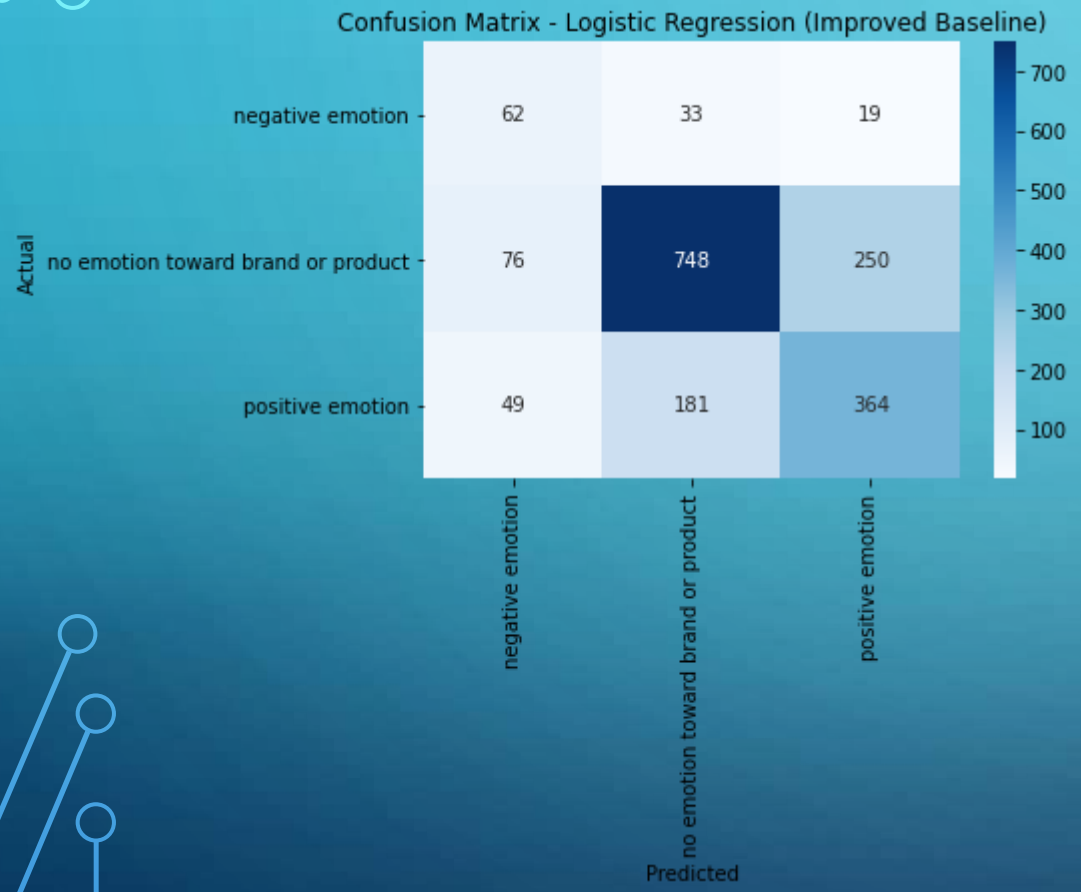
Negative sentiment is almost never detected

Slightly weaker performance than Logistic Regression

Insight:

This model is too conservative and not suitable for practical sentiment monitoring.

LOGISTIC REGRESSION (IMPROVED)



Interpretation:

Significant improvement in detecting negative tweets

More balanced classification across all classes

Increase in false positives but better coverage of real negatives

Insight:

This model achieves the best trade-off between accuracy and business relevance.

NAIVE BAYES (IMPROVED)

Interpretation:

Slight improvement compared to baseline

Still dominated by neutral predictions

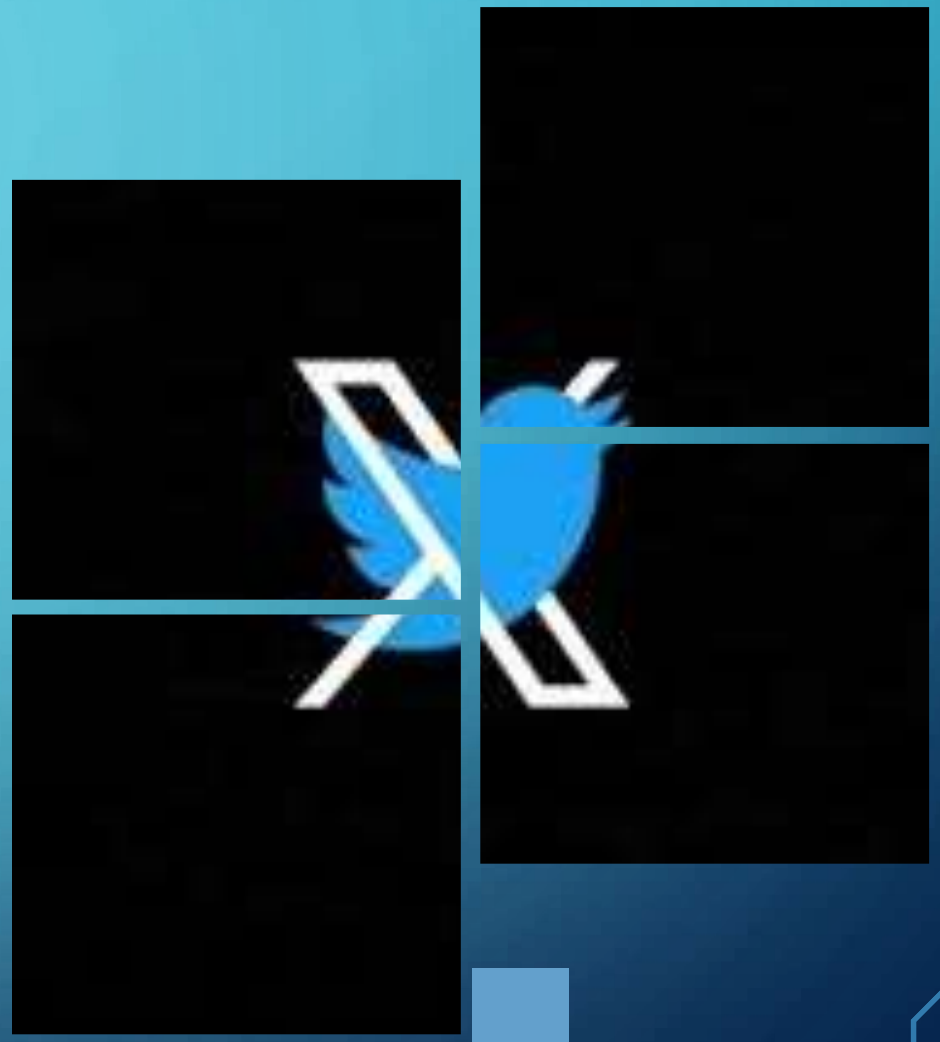
Negative detection remains weak

Insight:

Improvements are limited and do not solve the main problem.

MODEL COMPARISON

- The comparison shows that overall accuracy alone is not sufficient to evaluate model performance. Baseline models perform well on dominant classes but fail to detect critical negative sentiment.
- The improved Logistic Regression model stands out because it significantly increases negative recall while maintaining reasonable performance across other classes.
- This makes it the most practical model for real-world use.

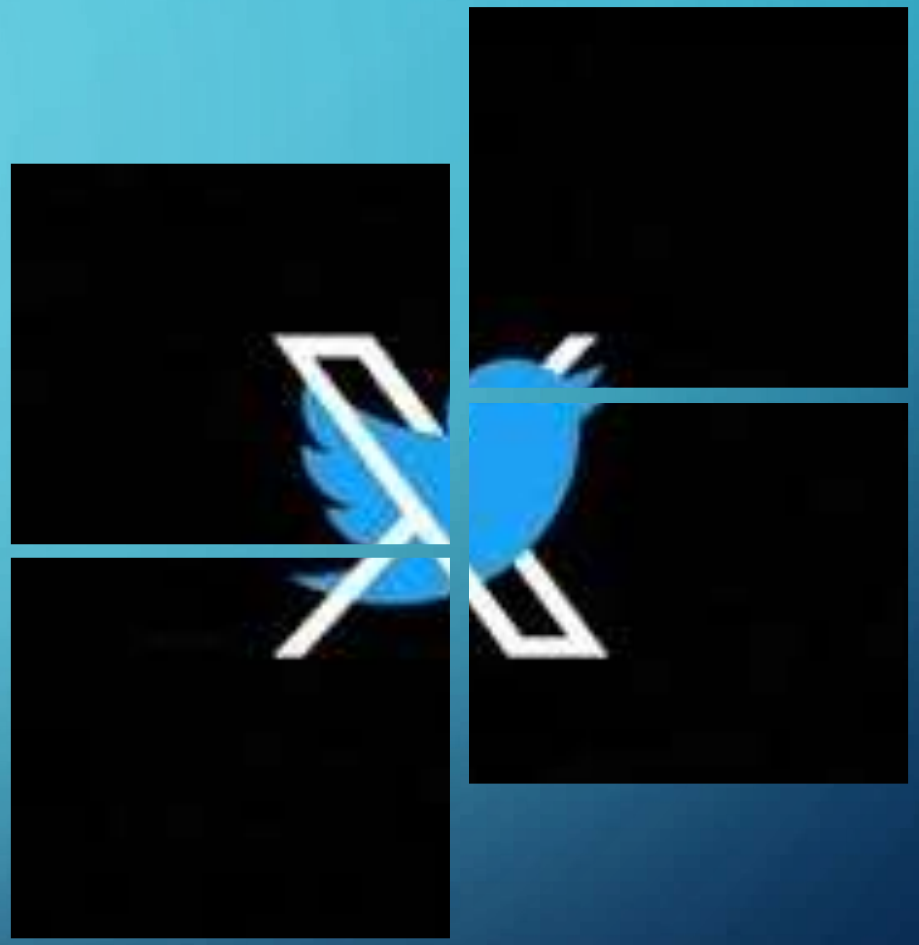


KEY INSIGHT

- The most important takeaway from this project is that **model evaluation must align with business goals**.
- In this case, detecting negative sentiment is more valuable than achieving high overall accuracy. A model that misses negative feedback can lead to poor business decisions, even if its accuracy appears high.
- Therefore, recall for negative sentiment is a key performance metric.

LIMITATIONS

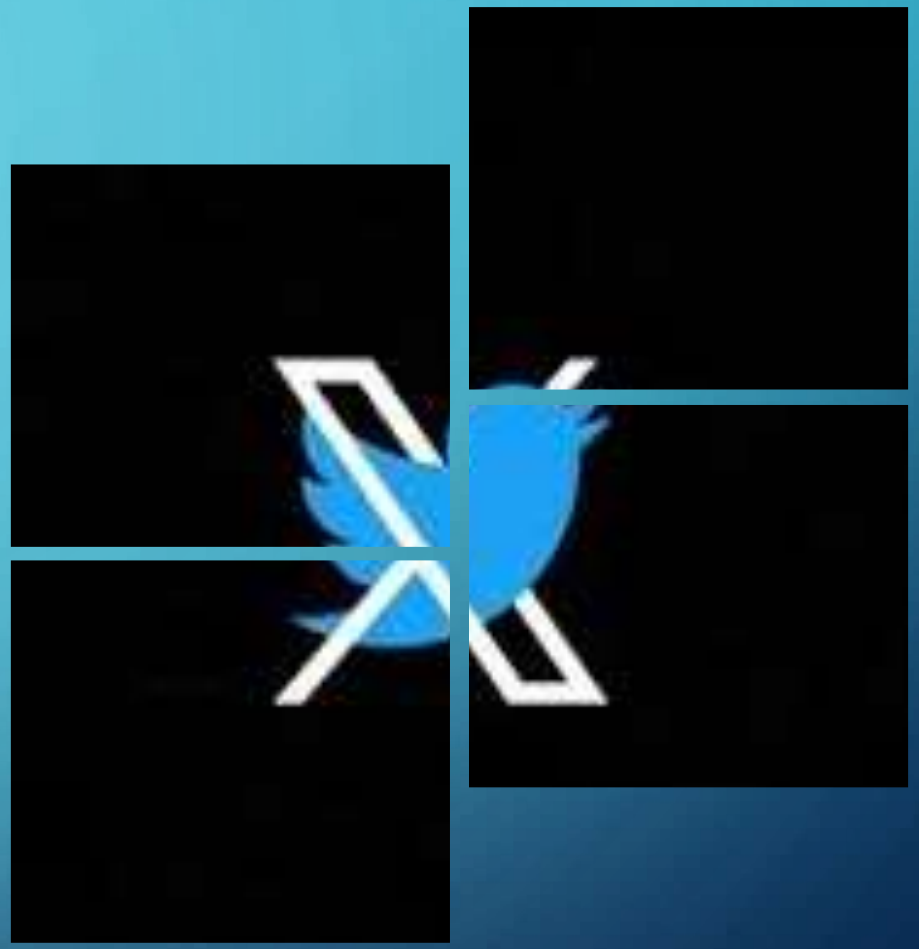
- The project is limited by class imbalance, which affects how models learn from data. Additionally, tweets often contain sarcasm and informal language, which are difficult to interpret using traditional NLP techniques.
- The use of TF-IDF also limits the model's ability to capture deeper context, as it focuses on word frequency rather than meaning.
- These limitations highlight areas for improvement rather than weaknesses.



FUTURE IMPROVEMENTS

- Traditional models are limited in capturing context and handling imbalanced data.
- Advanced NLP models such as BERT can better understand sentence structure and improve sentiment classification.
- Increasing dataset size and balancing classes would further enhance model performance.
- Deploying the model in a real-time system enables continuous monitoring of customer sentiment.

This transitions the project from analysis to a practical business solution.



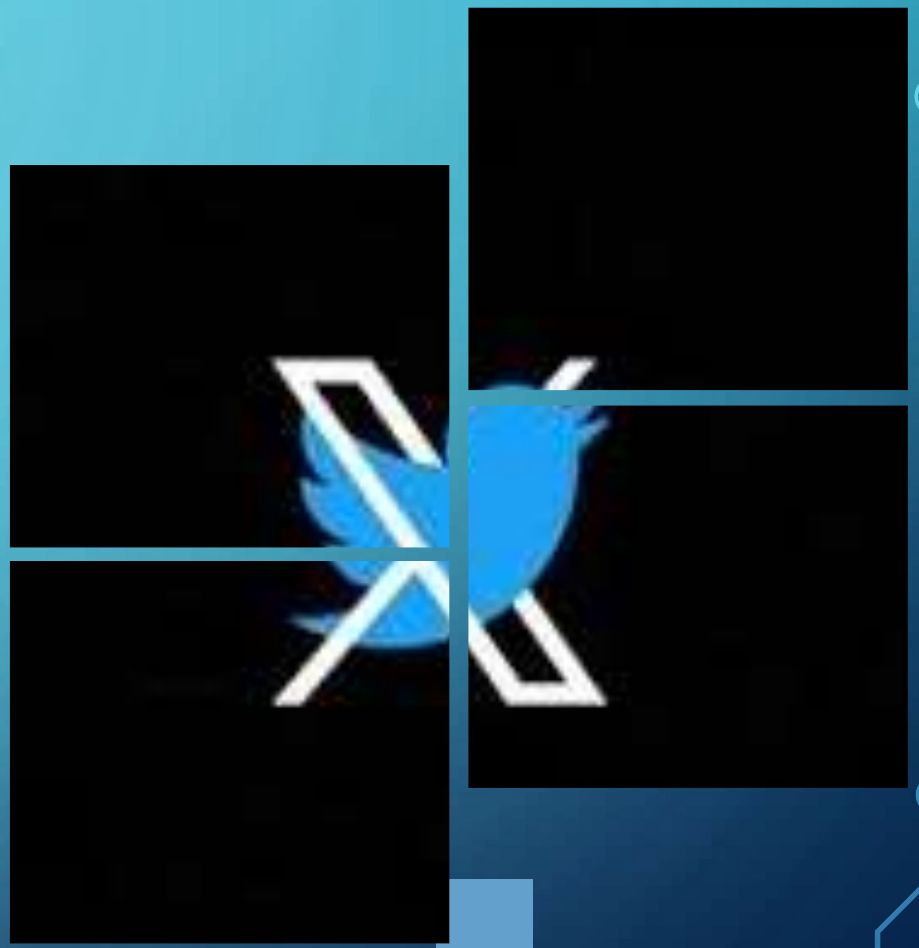
DEPLOYMENT MODEL: TRANSFORMER-BASED APPROACH (BERT)

A transformer-based model was deployed to overcome the limitations of traditional approaches. Unlike word-frequency models, it analyzes full sentence structure to better capture meaning.

Key Points

- Context-Aware Understanding: Reads text bidirectionally, capturing full sentence meaning
- Handling Complex Language: Improves detection of negation, informal phrasing, and subtle sentiment
- Transfer Learning Advantage: Uses pre-trained language knowledge for higher accuracy

Key Insight: **Transformer models capture context, making them significantly more effective for real-world sentiment analysis.**



MODEL PERFORMANCE: TRANSFORMER-BASED (BERT)

The transformer-based model demonstrates strong overall performance, particularly in identifying positive and neutral sentiment.

Performance is evaluated using precision, recall, and F1-score, which provide a balanced measure of classification quality across different sentiment classes.

Interpretation:

- Positive sentiment achieves the highest performance, indicating strong model confidence
- Neutral sentiment is also well classified due to its dominance in the dataset
- Negative sentiment shows lower recall, reflecting the challenge of detecting minority classes

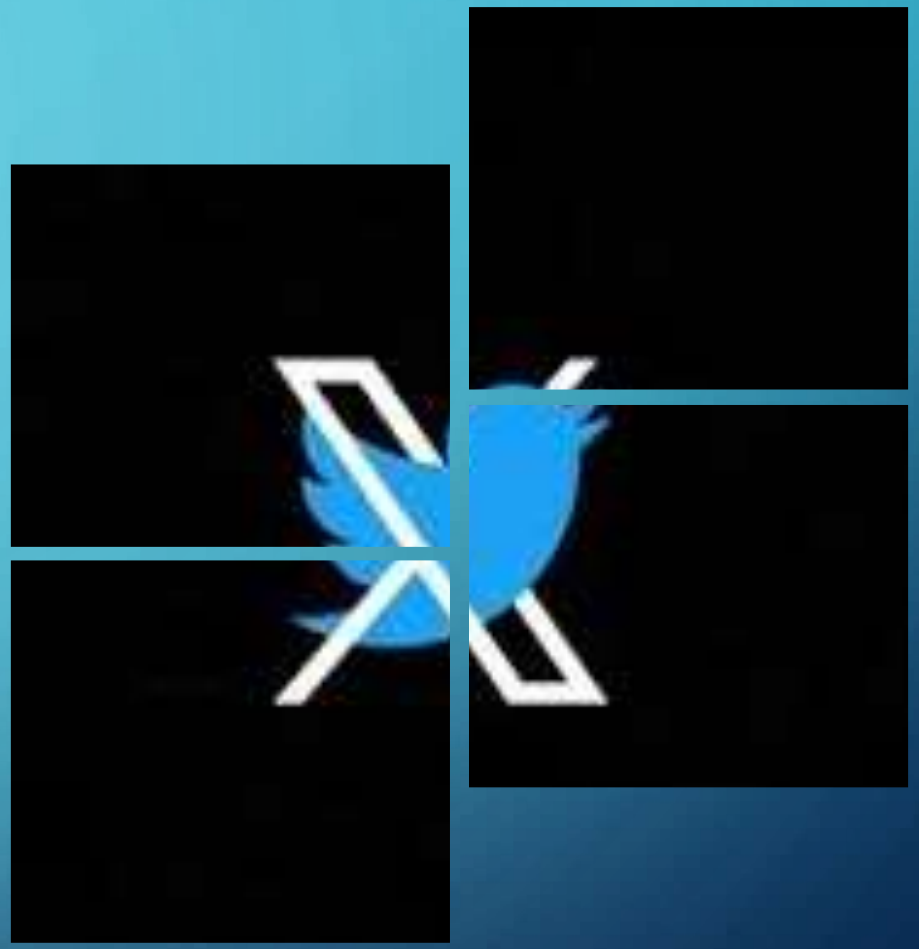
Classification Report	Precision	Recall	F1-score
Negative Sentiment	0.56	0.46	0.51
Neutral Sentiment	0.62	0.66	0.64
Positive Sentiment	0.78	0.76	0.77

FINAL INSIGHT & RECOMMENDATION

- The results show that traditional machine learning models struggle to detect negative sentiment due to class imbalance and limited contextual understanding.
- The transformer-based model significantly improves overall performance by capturing sentence structure and meaning more effectively.
- However, detecting negative sentiment remains the most challenging aspect, highlighting the need for further optimization and balanced data. This transitions the project from analysis to a practical business solution.

Key Recommendation:

A transformer-based approach is the most suitable for real-world deployment, but model performance should be continuously improved with better data balancing and advanced tuning.



CONCLUSION

This project demonstrates how NLP can be used to analyze large volumes of unstructured tweet data and extract meaningful insights.

While multiple models were tested, the improved Logistic Regression model provided the best balance between performance and business relevance.

The key outcome is a system capable of identifying customer sentiment, particularly negative feedback, in a scalable and automated way.